Historical source

Encyclopaedia Britannica scan, source row 1910. Parse 3.5 extracted the text. Super 3.5 VL corrected it from an enlarged crop of each block. It can still contain recognition errors. Page headers, footers, and page numbers are removed.

Untitled continuation (E0167)

Continues an entry from an earlier page.

Lightning. we have seen above; it is mathematically evident, that, when a returning stroke comes to be produced by the sudden removal of the very strong elastic electrical pre- sure of the electrical atmosphere of a thunder-cloud powerfully charged; the strength of such a returning stroke must be enormous.

If indeed the quantity of electric fluid naturally contained in the body of a man, for instance, were immense, or indefinite, the author's estimate between the effects producible by a cloud, and those caused by a prime conductor, might be admitted. But surely an electrified cloud,how great soever may be its extent, and the height of its charge, when compared with the extent and charge of a prime conductorcannot expel from a man's body (or any other body) more than the natural quantity of electricity which it contains. On the sudden removal, therefore, of the pressure by which this natural quantity had been expelled, in consequence of the explosion of the cloud into the earth; no more (at the utmost) than his whole natural flock of electricity can re-enter his body (c). But we have no reason to suppose that this quantity is so great, as that its sudden re-entrance into his body should destroy or even injure him.

In the experiment above described, the inflated person receives into his body, at the instant of the returning stroke, not only all that portion of his own natural electric fire which had been expelled from it; but likewise transmits through it, at the same instant, in consequence of his peculiar situation, all the electric fire of which the large second conductor had been robbed; and which must necessarily re-pass through his body, to arrive at that conductor. To render the case somewhat parallel, in natural electricity, the man's body must be so peculiarly circumstanced, supposing him to be in a house, that the electric matter which has been expelled from the house into the earth, by the pressure of an extensive thunder-cloud, could not return back into the building, on the explosion of the cloud at a distance, without passing through his body: a case not likely to happen, unless the house were insulated (like the second conductor in the preceding experiment), and his body became the channel through which alone the house could have its electric matter restored to it: it appears much more probable that the electric matter returns to the house through the same channels by which it before insensibly passed out, and with equal silence, too, more suddenly.

and with equal evidence, the more indubitably. " In the case of a man who is abroad, and in an open field, during the time of an explosion;as he is unconnected with other masses of matter above him, no more than the precise quantity of electric fire, which had been before expelled from his body, will suddenly return into it at the instant of a distant explosion: and that this quantity is not usually very large, may be inferred from many considerations.

(c) " We suppose the perfor not to be fo situated, that the returning fire of other bodies mutt necessarily pass through his body."

(b) " The author does indeed produce a living evidence, in the case of a person at Vienna, who, he has been credibly informed, received an electrical shock, by having held one hand accidentally in contact with an intercepted metallic conducting rod, at the instant that a thunder-cloud exploded at the distance, as was conjectured, of above half an English mile. He likewise observes, that a very strong, bright, and sudden fire (or spark) of electric fire has been seen, by several electricians, to pass in the interval, or interruption, purposely left in the conducting rod of a house at the instant of a distant explosion; and "when it was fully proved, by the sharp point of the conductor not being melted, or even tinged," that the conductor itself had not been struck.Other observations, however, do not by any means prove the magnitude or danger, of the returning stroke, but merely its existence, which we do not contest.

When a person standing on the ground holds a pair of Mr Canton's balls in his hand, while a highly charged thunder-cloud is suspended over his head; the angle made by the balls indicates the electrical state of that person, or the quantity of natural electricity of which his body is at that time deprived, by the action of the (positively) charged cloud hanging over him. But we have never seen the repulsion of the balls to be considerable, as to furnish any just apprehensions that the return of his natural electric matter, however sudden, could be attended with injury to him: nor would he be sensible of any commotion on the balls suddenly coming together; tho' a spark might undoubtedly be perceived, at that instant, were he insulated, and placed in the same manner with the author when he tried the above-related experiment.

The author nevertheless observes, that there have been instances of persons who have been killed by natural electricity, having been found with their shoes torn, and with their feet damaged by the electrical fire; but who have not had, over their whole body, any other apparent marks of having been struck with lightning. He adds, "if a man walking out of doors were to be killed by a returning stroke, the electrical fire would rush into that man's body through his feet, and his feet only; which would not be the case, were he to be killed by any main stroke of explosion, either positive or negative."

It would be no difficult talk, we think, to account for these appearances in a different manner; were all the circumstances attending the case minutely ascertained: but without interrogating the dead on this subject, we may more satisfactorily appeal to the experience of the living (a), to shew, that though the returning stroke must take place, in all thunder-storms, in some degree or other; yet it is not of that alarming magnitude which the author ascribes to it. If, in any particular thunder-storm, a man in the open fields could be killed, at the instant of a distant explosion, merely by the return of his own electric fire, which had before been driven out of his body; surely numerous observations of persons who had experienced the returning stroke, in slighter degrees, would be familiar; and scarce a great thunder-storm must have occurred, in which one person or another must not, at the instant of an explosion, have felt the effects of the returning stroke, in some degree or otherfrom that of a violent concussion, to that of a slight and almost imperceptible pulsation. But no observations of this kind are known to us; nor have we ever heard of any person's experiencing any kind of electrical commotion in a thunder-storm, except such as have either been directly struck, or have happened to be in the very near neighbourhood of the spot where the explosion took place.

ook place. "The author has been aware of this objection, which

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